



**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Pediatric Neurology I

Principles and Practices of Osteopathic Medicine 1

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Do.
Make.
Heal.

Innovate.
Reinvent the Future.



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Session Objectives

- Understand general concepts of a neurological evaluation in pediatrics as well as the clinical presentation of seizures and initial management
- Understand criteria to differentiate b/w simple and complex febrile seizures
- Discuss other seizure types- characteristics and prognosis (absence, benign partial with centrotemporal spikes, infantile spasms, juvenile myotonic and Lennox-Gastaut)
- Understand non seizure conditions causing abnormal movements or events
- Understand various conditions causing ataxia

Define Seizures....

- “A seizure represents the clinical expression of abnormal, excessive, synchronous discharges of neurons residing primarily in the cerebral cortex. This abnormal paroxysmal activity is intermittent and usually self-limited, lasting seconds to a few minutes”

May not look like a “typical” seizure

- Children
 - Blank stares, lack of awareness
 - Falling
 - Unusual posture
- Infants
 - Lip smacking
 - Startles, jerks
 - Eye blinking, eye rolling
 - Repetitive movements
 - Abnormal breathing
 - Shudder/shiver

Causes and pertinent history

- Neonatal Seizures
 - Hypoglycemia
 - Hypoxic insult
 - Electrolyte Imbalances (Na,Mg,Ca, Phos)
 - Sepsis/ Meningitis
 - Hemorrhage
 - Metabolic
- Infants
- Older children
- What happened exactly? Describe details
- Time? How long it lasted
- What happened afterwards?



**DON'T PUT
ANYTHING
IN MOUTH**



**LOOK FOR
MEDICAL ALERT
IDENTIFICATION**

**TIME THE
SEIZURE WITH
A WATCH**

**DON'T
HOLD
DOWN**

**AS SEIZURE ENDS,
OFFER HELP**



**CUSHION HEAD,
REMOVE GLASSES**

**LOOSEN TIGHT
CLOTHING**

**TURN
ON SIDE**

-
- A 15 month old male no PMHx presents to the ER via ambulance after having a seizure which lasted less than 5 minutes, Shaking all over, eyes rolled to the back of his head. After it stopped he went to sleep. He has had a mild cold last 2 days but had no fever. While in the ER, temp 103 degrees, woke up and now alert and responsive.

Febrile Seizures

- Criteria: (simple)
 - A convulsion associated with an elevated temperature greater than 38°C (100.4°F) (rate of rise)
 - 6 month to 5 years of age
 - Lasts less than 15 minutes
 - Tonic/Clonic
 - Post-ictal phase after
 - No prior neurologic issue
 - No central nervous system infection or inflammation
 - No acute systemic metabolic abnormality that may produce convulsions
 - No history of previous afebrile seizures

Febrile Seizures

Simple (Benign)

- Generalized tonic-clonic
- Less than 15 min
- Age range 6months-5yrs
- One seizure in a 24hr period
- Full recovery after post-ictal phase
- **Usually reassurance and no further workup**

Complex

- Focal features
- Longer than 15 minutes
- Outside age range
- More than one episode in 24 hrs
- Residual symptoms or paresis after post-ictal phase
- Pre-existing neurological issues

Simple Vs Complex Febrile Seizure

- Management?
- Specialist referral?
- Recurrence?
- Future epilepsy?
- Testing?
 - EEG
 - MRI
 - CT scan



Febrile Seizure and Epilepsy

- Etiology of febrile seizures is unclear
- **Recurrence rate varies: 30-65%, the more recurrences, the more likely to have more episodes**
- Genetic susceptibility: among first-degree relatives of children with febrile seizures, 10 to 20 percent of parents and siblings also have had or will have febrile seizures
- Treatment: recommended if the febrile seizure lasts longer than **5 minutes**; options include lorazepam IV, fosphenytoin IV, **rectal diazepam gel**
- Neurologic sequelae are rare
- Subsequent epilepsy frequency only slightly above general population

Infantile Spasms AKA West Syndrome

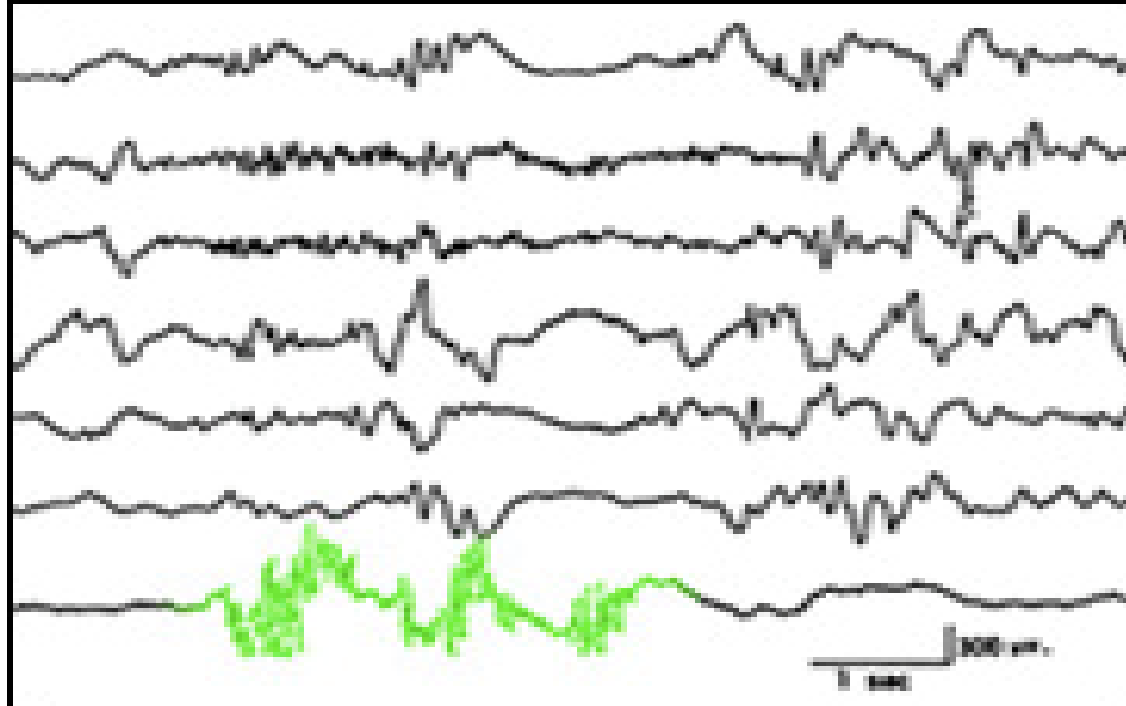
- Characterized by epileptic spasms with onset in infancy or early childhood that are usually associated with
 - Electroencephalographic (EEG) pattern of **hypsarrhythmia (chaotic)**
 - **Developmental regression (loss of milestones)**

Infantile Spasms/ West Syndrome

- The majority of affected children present at **less than one year of age** (range from one day to 4.5 years of age)
- Peak incidence of onset is between 3-7 months
- True age of onset may be uncertain because may be initially mistaken for other conditions
- **Majority of children will go on to be cognitively impaired and/or develop other seizure disorders**

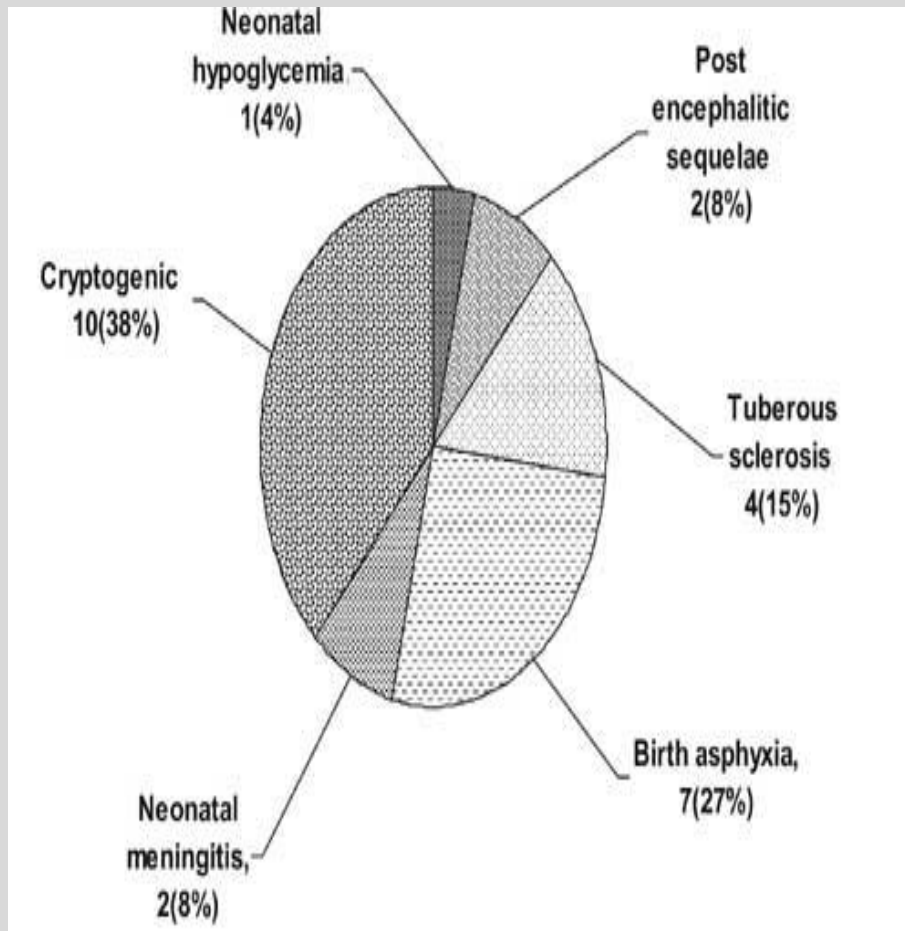
Hypsarrhythmia

EEG of a 5-month-old infant with infantile spasms showing typical hypsarrhythmic pattern **in GREEN**



Infantile Spasms

- Causes...



- Tuberous sclerosis- Ash leaf spot



Infantile Spasms continued..

- Treatment
- Hormonal therapy such as **Adrenocorticotrophic hormone (ACTH)**, and antiepileptic medication, mainly **vigabatrin**
- Many questions still remain regarding the mechanism, optimal drug, dose, duration of therapy, and the importance of prompt initiation of treatment after the appearance of spasms

Benign myoclonus of infancy

- Also known as benign non-epileptic infantile spasms
- Myoclonus, spasms with brief tonic contractions or shuddering
- First occurring at three to eight months of age, these clusters increase in intensity and severity over weeks or months and then remit spontaneously at two to three years of age
- Typically occur when awake
- Spasms can occur in clusters
- **Normal EEG**, normal neurologic development without subsequent epilepsy
- **Reassurance** (after confident it's not infantile spasms or other seizures)

Absence Seizures

- Sudden, profound impairment of consciousness without loss of body tone characteristically provoked by hyperventilation
- **May be subtle** and often go unnoticed for prolonged periods, even with multiple seizures per day (**often confused with inattention/daydreaming**)
- Median age of onset: 6 years old (Range 4-10)
- Often remits by early puberty in most patients without major sequelae
- Up to 20 percent of children have a history of febrile seizures and nearly half have a first or second degree relative with seizures

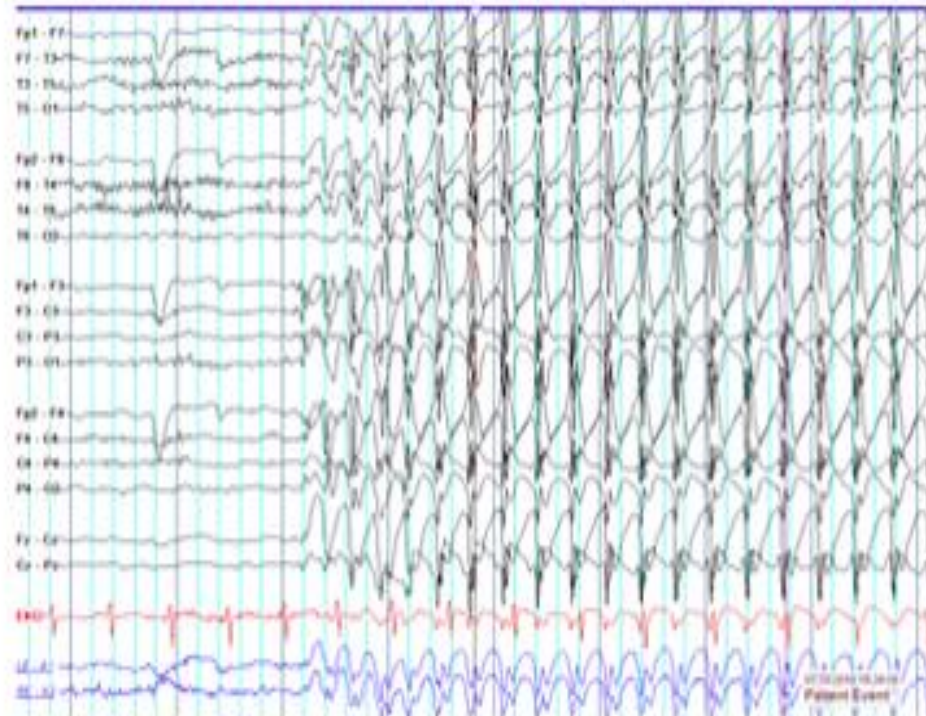


Absence Sz –Diagnosis

- Requires all of the following:
 - Age at onset of 4 to 10 years
 - Normal neurological and developmental state
 - Brief (4 to 20 seconds) and frequent (tens per day) absence seizures, with **abrupt and loss/ alterations of consciousness**
 - Generalized rhythmic spike and wave discharges at around **3 Hz (EEG)**

Need High index of suspicion!

Absence Seizure



Treatment

- Ethosuximide (first line therapy), Valproic acid
- Several antiepileptic drugs have the potential to aggravate absence seizures and should be avoided (Carbamazepine, Gabapentin, Phenytoin)

Benign partial epilepsy with central temporal spikes (BECTS)

AKA **Rolandic** Epilepsy

- Most common seizure type is a **simple partial** seizure with motor symptoms involving the face
- Age of onset peaks between 7-9 (Range 3-13)
- $\frac{3}{4}$ of seizures occur at night (**falling asleep, asleep, or upon awakening**)
- Electroencephalogram (EEG) findings of BECTS are characteristic, with **centrotemporal** sharp waves (70 to 100 msec) that have several distinctive features

BECTS continued..

- **Excellent prognosis**, with spontaneous remission in the majority of patients before age 12 to 13 years
- Treatment: Antiepileptic drug (AED) treatment is often not recommended, particularly if only simple partial seizures occur and if the child and family are comfortable with this approach

Juvenile Myoclonic epilepsy

- Underlying cause unknown
- Typically occurs in otherwise healthy **adolescents**
- Triad: Myoclonic jerks, generalized tonic-clonic seizures and absence seizures
- **Photosensitivity is common**
- **Seizures occur upon awakening or in association with sleep deprivation**

Juvenile myoclonic epilepsy continued..

- Most patients with JME have **normal** global cognitive capacities
- Diagnosis: Interictal EEG pattern shows **4-6 Hz** bilateral polyspikes and slow wave discharges with frontal predominance over a normal background activity

Juvenile Myoclonic epilepsy continued..

- Treatment: generally respond quickly and completely to broad spectrum antiepileptic drug (AED) therapy, but **most require lifelong treatment**
- Valproic acid is first line (controls all 3 seizure types), Lamotrigine
- Lifestyle modifications- sleep, hydration, avoid flickering lights

Lennox-Gestaut Syndrome

- Associated with **severe seizures** in childhood
- Many causes: Tuberous sclerosis, meningitis, head injuries
- Usually present in the first 7 years of life
- Up to 25% have a history of infantile spasms
- Clinical features characterized by:
 - Multiple seizure types**
 - A slow (less than 2.5 Hertz) spike-wave pattern on the interictal EEG that is generalized and of highest voltage in the frontal region (also called an "atypical spike and wave" pattern)
 - Cognitive dysfunction (occasionally progressive), psychotic symptoms

Lennox-Gestaut Syndrome

- Hard to control usually need more than one AED
 - Treatment- Valproate, Topiramate, Lamotrigine,
- The optimal therapy for LGS is uncertain, no drug has been shown to be highly effective
- Ketogenic Diet

SYNDROME	CLINICAL DESCRIPTION	EEG FINDINGS	TREATMENT	PROGNOSIS
Infantile spasms	Onset between 4 and 8 months of age. Seizures consist of repetitive volleys of brief flexion or extension contractions of the neck, trunk, and extremities that persist for 10 to 30 seconds per volley. Frequently, loss of developmental milestones with the onset of seizures.	High-voltage bilaterally asynchronous, and irregular high-voltage spike and wave pattern.	• Adrenocorticotrophic hormone (ACTH) • Benzodiazepines • Vigabatrin (infantile spasms and tuberous sclerosis) • Epilepsy surgery when focal onset	Guarded; majority have poor outcome with persistent seizures.
Lennox-Gastaut	Common in preschool children. A mixture of seizures, including myoclonic, generalized tonic-clonic, partial, absence, and atonic, is characteristic. Status epilepticus is frequent. Often occurs in children who previously had encephalopathy.	Abnormal background activity, slow spike-waves, and multifocal abnormalities.	• Valproic acid • Benzodiazepines • Ketogenic diet	Unfavorable; high association with mental retardation and behavioral problems
Benign childhood epilepsy with centrottemporal (rolandic) spikes	Peak onset between 9 and 10 years of age (range, 2 to 14 years). Majority occur during sleep. Child awakened by unilateral tonic-clonic contractions of the face, paresthesias of tongue and cheek, and occasional clonic seizures of ipsilateral upper extremity. Child is conscious but aphasic for several minutes.	Repetitive spike discharges confined to the centro-temporal area with normal background activity.	• Occasional seizures require no therapy • Frequent seizures controlled by carbamazepine	Excellent; spontaneous remission by adolescence.
Juvenile myoclonic epilepsy (Janz syndrome)	Onset between 12 and 16 years of age. Myoclonic jerks on awakening that diminish later in day. Most develop early morning generalized tonic-clonic seizures.	A 4 to 6 per second irregular spike and wave pattern enhanced by photic stimulation.	• Valproic acid	Excellent, but valproic acid is required life-long.

Abnormal Movements/Paroxysmal events

- “My child is acting funny”
- “Does weird movements”
- “I think he/she is having a seizure”



It's NOT a Seizure

- Chorea
- Dystonia
- Reflux
- Ataxia
- Breath Holding Spells

Chorea

- Dance-like, writhing, chaotic involuntary movements
- St Vitus' Dance
 - **Sydenham chorea- Acute Rheumatic Fever- Group A strep**
 - Lupus
 - Wilson's Disease
 - Cerebral Palsy
 - Lyme's Dz
 - Hyperthyroid

Dystonia

- Muscle contractions leading to twisting/ abnormal postures
- Similar to seizure
- Cerebral Palsy
- ***Acute Dystonic Reaction***
- Child given anti-emetics esp phenothiazines, metoclopramide, antipsychotics (haloperidol)
- **Give IV Diphenhydramine (Benadryl)**

Sandifer syndrome

- Esophageal Reflux
- Arching back
- Looks like a seizure
- Reflux precautions



Breath Holding Spells: Cyanotic vs Pallid

- **Cyanotic** (most Common)
 - Classic history
 - Tantrum, cries and not taking a breath, till turns pale/ blue, passes out. Occasionally twitches/ shakes like a seizure.
 - Most at 1-3 yrs old- tantrum age but can be 6mo to 6 yrs
 - **Reassure, check for Iron Def anemia** (CBC, ferritin), parenting advice



Pallid Type Breath holding Spell

- **Sudden event** after stimulus/ shock/ surprise/ minor trauma
- Pass out with color change/ cyanosis
- Easily confused with seizure more
- **Abnormal Vagal response-** may not completely outgrow
- **Reassure, check for Iron Def anemia** (CBC, ferritin), parenting advice

Self-Stimulation

- Rhythmic rocking, movements
- Orgasm may look like a seizure
- Young toddler and up
- Reassure, keep child active/ distracted/
change the environment

The Ataxic Child- Differential

- Acute Cerebellar Ataxia- Most common
- Benign positional Vertigo
- Guillan-Barre syndrome
- Trauma/ Intracranial Hemorrhage
- Medications/ Toxic ingestions
- Neuroblastoma
- Ataxia telangiectasia
- Friedrich's Ataxia
- Tumors- Posterior Fossa

Acute Cerebellar Ataxia

- 2-5 yrs old
- Post- infectious
- Self-limited
- Most common cause ...but diagnosis of exclusion
- Acute Labyrinthitis looks similar- OM/viral infection
 - Hearing loss, vomiting, vertigo

Benign Positional Vertigo

- 2-12 yrs, ave ~6yrs old
- Confused, sits down- doesn't want to stand, nystagmus
- Certain maneuvers cause/ exacerbate
- Short lived, seconds to minutes only
- Migraine Variant

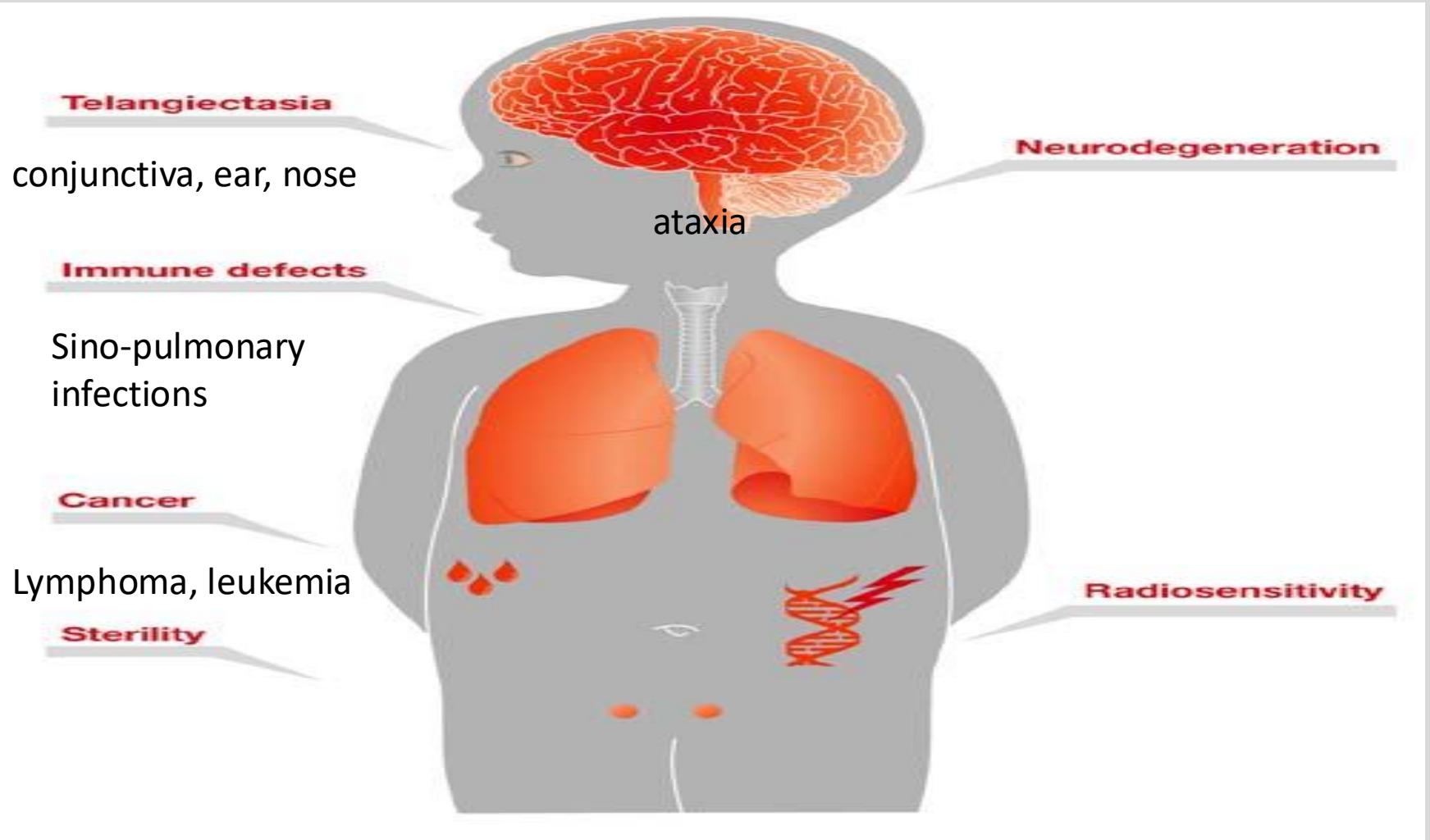


Ataxia telangiectasia

- Start age around 2 years, loss of ambulation by adolescence
- Sino-pulmonary issues
- Long term risk of Cancers
- Susceptible to chromosome breaks

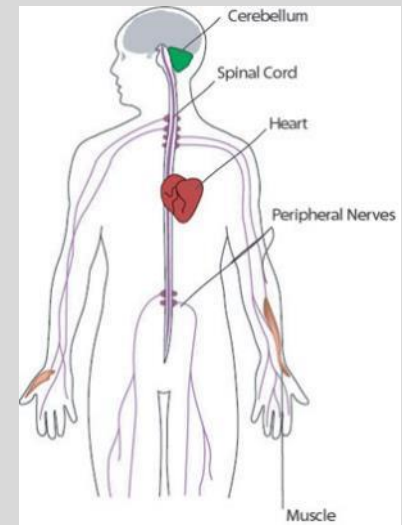


Ataxia telangiectasia



Friedreich's Ataxia

- **Ataxia**
- Distal muscles weak- legs worse
- Loss of DTR's esp Achille's
- High arched Feet/ Hammertoes
- **Hypertrophic Cardiomyopathy**
- **Later onset than Ataxia telangiectasia (10 yrs old or later)**



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- Add Practice Questions

Summary Slide- Part 1

- Seizures in infants and children can be subtle- blank stares, facial movements, posturing
- Simple febrile Seizures vs Complex. Simple usually no further workup but know the recurrence rate is high till outgrown
- Infantile Spasms/ West Syndrome: Infancy- Hypsarrhythmia, dev regression, poor prognosis. Assoc with Tuberous Sclerosis
- Benign Myoclonus of infancy- jerks in infancy, normal EEG, reassurance
- Absence Seizures- Blank stares/ inattention/ daydreaming in early elementary age student. 3 Hz spike and wave, Rx: ethosuximide
- Rolandic Epilepsy/ BECTS- simple partial around sleep/arousal, Centrotemporal waves, will usually outgrow
- Juvenile Myoclonic epilepsy- sleep deprivation or bright flashing lights can trigger, adolescents, controlled with Valproate. Usually lifelong tx but good prognosis
- Lennox-Gestaut- difficult to control, complex seizures, poor prognosis

Summary Slide- Part Deux

- Chorea- Writhing involuntary movements esp think Sydenham chorea from rheumatic fever
- Acute dystonic reactions- esp medication side effect, give IV diphenhydramine (Benadryl)
- Sandifer syndrome- reflux in infant looks like a seizure- reassure
- Breath holding Spells- child crying and holds breath till turns blue and passes out- reassure
- Self stim in children can look like a seizure- reassure, change the environment
- Acute Cerebellar Ataxia- Post infectious, self limited, diagnosis of exclusion
- Benign positional Vertigo- young child looks confused, unbalanced
- Ataxia Telangiectasia- young childhood, telangiectasias, recurrent infections, risk of cancers
- Friedrichs Ataxia- distal weakness and loss of reflexes, cardiomyopathy, foot issues

Lecture Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>



To be continued...